

Software Engineering

5430

Spring 2023



by:

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Computer Science Department

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Discovery Park F203

Agenda

- **Why is COMP-5430?**
 - **Course Description & Goals**
- **Course Outline**
 - **Topics**
 - **Lectures**
 - **TAs/IAs**
 - **Textbook**
 - **Evaluation**
 - **Instructor**
 - **Rules**



Course Overview

Course Description

Case tools, module implementation, testing, system delivery in the work place, scheduling and budgeting, project management, configuration management, software development tasks and ethical issues.

What is 5430?

- This course introduces the fundamental concepts, common principles, and general techniques of software engineering. It discusses the main issues involved in the development life-cycle of nontrivial software systems, including process models, feasibility studies, requirements elicitation and definition, rapid prototyping, design methodologies, verification and validation, and software evolution.
- Students taking this course are required to work on projects, which are designed to go through the major phases of large-scale software system development.

Course Objective & Goals

- Learn students to use the best practices in software engineering.
- Upon completion of the course, you will be able to:
 - Explain the various software process models
 - Apply requirements elicitation, specification, and validation processes
 - Use the Unified Modeling Language (UML) to represent the design models.
 - Employ methods for software testing
 - Develop a system based on software quality standards and guideline

Course Objective & Goals ...

- Upon completion of the course, you will be able to:
 - Apply an iterative, use case-driven methods to develop robust design models.
 - Describe the different views of software architecture.
 - Implement the basic concepts of the software design patterns.
 - Apply Object-Oriented (OO) concepts: abstraction, encapsulation, inheritance, hierarchy and polymorphism to implement these models.
 - Software project management and quality assurance.

TAs/IAs

Evaluation



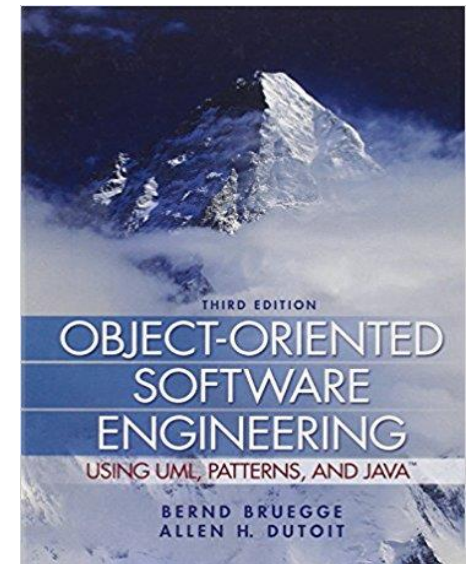
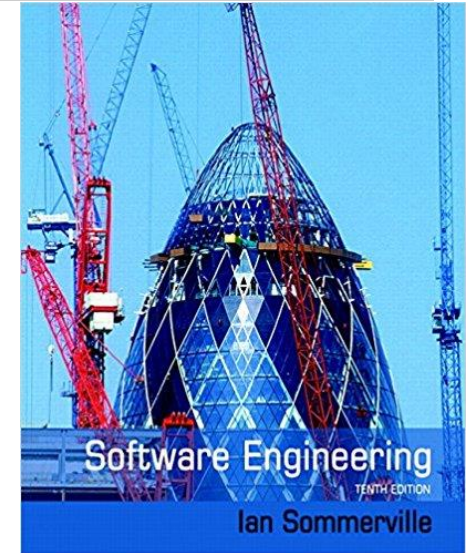
Course Component	Weight	Descriptions
Project0 - Team Charter	2%	
Project1 - SRS	24%	in 3 Phases
Project2 - Implementation	18%	
Project3 - Testing	16%	2 parts
Exam1	15%	Closed Book
Exam2	25%	Closed Book

Course Works Submissions

- Project's Submission:
 - Project reports are required to be submitted in the course Canvas site by the prescribed due date.
 - Late projects:
 - Accepted up to three days after the deadlines.
 - Weekend day count as a single day.
 - Late penalty of 10% of the available marks per day.
- Plagiarism:
 - You should read the definition and penalties of scholastic offense policy in the University of North Texas Academic Calendar at: <http://www.unt.edu/-----.html>

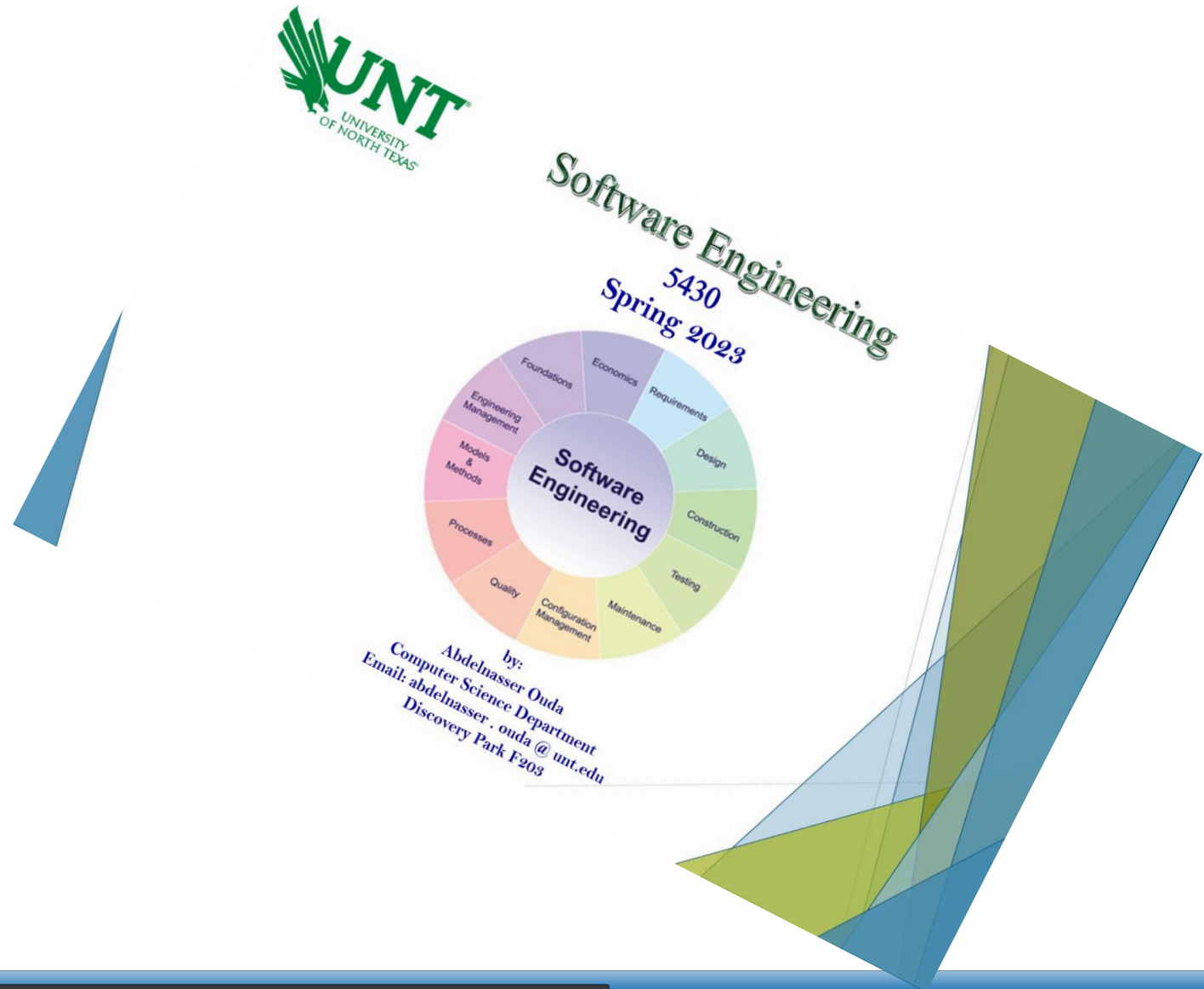
Course Textbook (s)

- Title: Software Engineering
 - Author: Ian Sommerville
 - Publisher: Pearson
 - Edition: 10th or newer
 - ISBN: 0-13-394303-8
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- Title: Object-Oriented Software Engineering
 - Author(s): Bernd Bruegge & Allen H. Dutoit
 - Publisher: Addison-Wesley
 - Edition: 3rd or newer
 - ISBN: 0-13-606125-7



Course Website

- The course website is an integral part of this course
- All course notes will be available there
- All project submissions are through the website
- All grades (except final exam and final grade) will be available there
- It is the student's responsibility to check the course announcements.



IMPORTANT NOTES

- Attending classes is part of the course components. Students must note that the Canvas content is certainly not a substitute for attending lectures, completing assigned readings, programming and general practice.
- Students must not rely on previous examinations as sole study aids – examinations are created anew for each semester.
- Finally, Canvas will be used for some announcements, but students are expected to attend lectures to stay fully informed about course related matter and materials.

Instructor

- Abdelnasser Hassan Ouda
- Office: Discovery Park F203
- Email: [abdelnasser . ouda @ unt.edu](mailto:abdelnasser.ouda@unt.edu)

Instructor - Education

- **Ph.D.
Computer Science**
Date: September 2009
Institution: University of Western Ontario, London, Ontario, Canada
Research
 - Distributed SystemsFocus:
 - Autonomic Computing
 - Security
 - Software Engineering
 - Self-Configuring Management Systems
 - Agents Deployment Models & Events Correlations
 - Database Management
- **Master's of Science
Computer Science**
Date: August 1993
Institution: Ain Shams University, Cairo, Egypt
Research
 - Access Controls PoliciesFocus:
 - Cryptography
- **Bachelor's of Science
Pure Mathematics and
Computer Science**
Date: May 1986
Institution: Ain Shams University, Cairo, Egypt
Grade: 96.80% (with the highest honour in Ain Shams University)

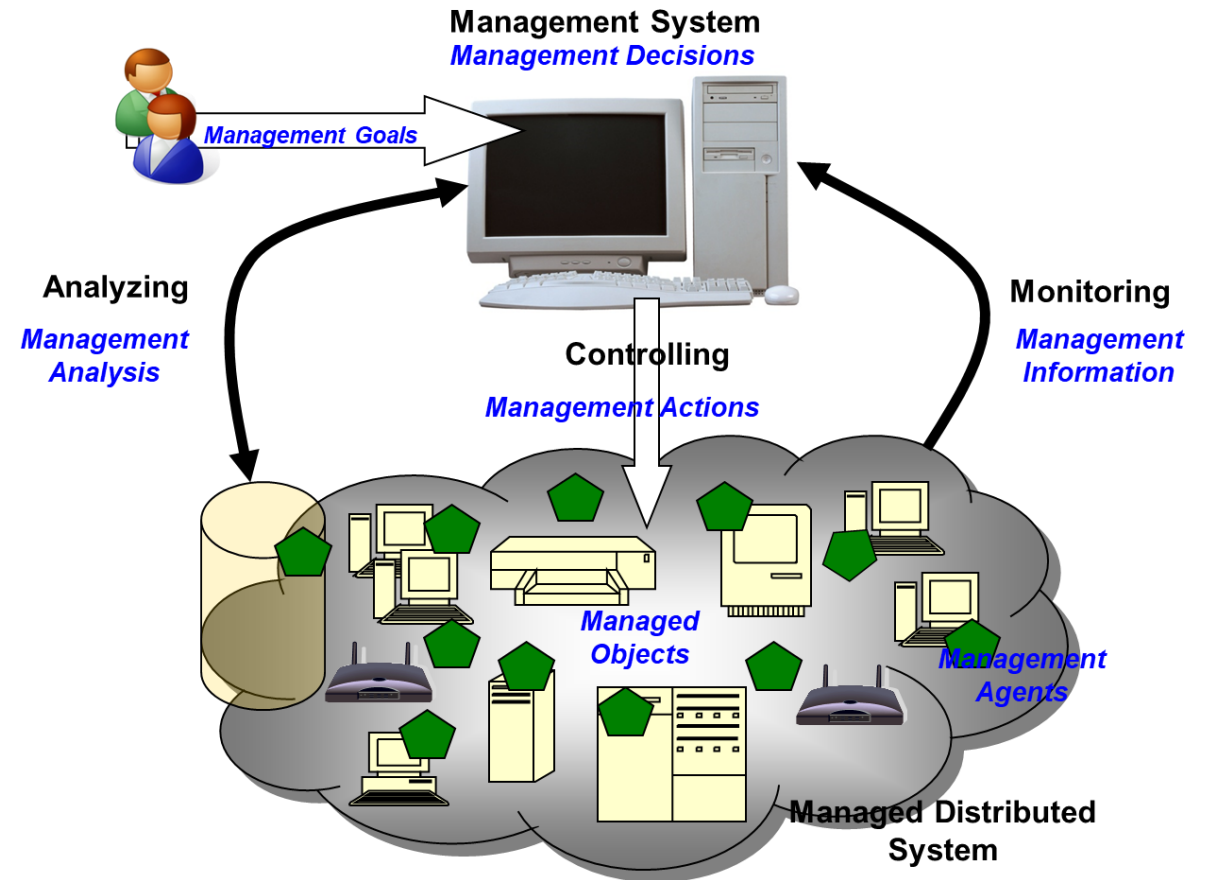
Instructor – Research Interests

- Distributed Systems, Distributed Systems Management and Parallel Programming
- Autonomic Computing
- Agents and Middleware Technologies
- Networking and Computer & Network Security
- Cyber Security and Cryptography
- Software Engineering and Database Management
- Embedded systems
- Clouds and Peer-to-Peer Computing and Visualization
- Mobile Systems

Instructor Research Interests -Autonomic Computing

Towards Automating Management Systems: Self-* capabilities

- Self-Configuring
- Self-Protecting
- Self-Healing
- Self-Optimizing



Instructor – Employment History

• Sept 2021–Present	Clinical Associate Professor	University of North Texas, Texas, US
• Jan 2020–Aug 2021	Instructor (Computer Science Department)	Western University, Ontario, Canada
• Sept 2019–Dec 2019	Sessional (School of Computer Science)	Windsor University, Ontario, Canada
• July 2017–June 2019	Assistant Professor (two years contract)	Windsor University, Ontario, Canada
• Sept 2015–Aug 2016	Assistant Professor (one-year contract)	Western University, Ontario, Canada
• Sept 2013–May 2015	Associate Faculty / Teaching	Lambton College Ontario, Canada
• Sept 2000–Aug 2013	Researcher/Teacher Assistant/PMagic Software Architect, Developer & Owner/ PhD Candidate	Western University, Ontario, Canada
• Nov 1995–Aug 2000	Senior Project Manager/Development Section Head/Solution Architect/Database & UNIX Administrator/ Senior Project Manager/Development Section Head/Solution Architect/Database & UNIX Administrator/Network Engineer/ Lecturer	Ministry of Interior, UAE
• May 1997–Sept 2000	Project Owner/Leader/System Analyst/ Programmer/Network Engineer /Database & MSNT Administrator/Network Engineer	Al-OudaSoft Project UAE
• Jan 1988–Oct 1995	Financial Automation Group Chair/Project Leader/ Senior System Analyst/Programmer/Teacher	Information Technology Center- UAE University, UAE
• Nov 1986–Dec 1987	Project Manager/Analyst/Programmer/Teacher	Al-Ain Computer Center, UAE
• Sept 1987–Aug 1988	Computer Science Teacher Grades 4-12	Al-AIN Scientific School, UAE
• Jun 1984–Oct 1986	System Analyst/ Programmer/Teacher	National Group Company, Cairo, Egypt

General Information – Abdelnasser Ouda

- (1) Ph.D. Computer Science, CSD, UWO
- (2) Researcher/Instructor Computer Science, UWO
- (3) Associate Professor, Lambton College
- (4) Assistant Professor, ECE, UWO
- (5) Assistant Professor, CS, Windsor University
- (6) Instructor, CS, UWO

B.Sc. Computer Science, Pure Math. M.Sc. On Computer Sec.
Ain-Shams University, Cairo, Egypt



16 years industry experience: Software engineer, network engineer, and project manager. U.A.E.

Clinical Associate Professor, CSCE, UNT

- Over than 23 years teaching experience
- Over than 35 years IT industry experience

Instructor – Hobbies

- TAEKWONDO player and instructor with approved 3rd degree black belt from Kukkiwon/World Taekwondo Federation in August 1983.
- Won several Gold and Silver Medals in Taekwondo tournaments beside the trophy of the Best Coach twice
- Love running and swimming
- Playing Drums and Chess.
- Enjoy reading and writing Arabic poem.

Questions?

